
When Does Retrieval-Augmented Generation Help?

Omar Jallouq
Bocconi University
omar.jallouq@studbocconi.it

Filip Janczak
Bocconi University
filip.janczak@studbocconi.it

Mohammad Aaqib Khatibi
Bocconi University
mohammad.khatibi@studbocconi.it

Dhruv Singh
Bocconi University
dhruv.singh@studbocconi.it

Abstract

Retrieval-Augmented Generation augments language models with non-parametric external memory, but its benefits depend on several factors that must be studied in controlled isolation. We investigate the area of benefit of retrieval through ablation on TriviaQA ($n = 750$ questions, pooled Wikipedia/web corpus) using Flan-T5-base as the generator, using 12 experiments to cover retrieval method, chunk granularity, retrieval depth, prompt template, oracle and cross-encoder re-ranking upper bounds, external distractor robustness, controlled distraction at fixed recall, forced retrieval miss, and generator scale. Our main findings are: (i) RAG provides a massive benefit over the parametric baseline ($\Delta\text{EM} = +0.551$, Holm $p < 10^{-4}$) because Flan-T5-base genuinely lacks long-tail factual knowledge; (ii) the generator, not the retriever, is the binding constraint at this scale as retrieval method, embedding model, and cross-encoder re-ranking are all statistically indistinguishable under Holm correction; (iii) with the answer guaranteed present, each additional hard topical distractor monotonically reduces EM ($0.673 \rightarrow 0.558$ across $N = 0-8$ distractors); and (iv) wrong, but topical, retrieved context primes parametric recall ($+0.102$ EM over an empty-context baseline), so a retrieval miss is costly relative to correct retrieval but does not drag the model below its parametric floor.

1 Introduction

Large language models encode substantial world knowledge during pre-training, but this parametric memory has well-known failure modes: facts can be outdated, hallucinated, or simply absent for long-tail entities [1]. Retrieval-Augmented Generation (RAG) addresses this by augmenting the generator with a non-parametric corpus, retrieving relevant passages at inference time and conditioning the model on them.

Despite RAG’s strong empirical track record, its benefits are not uniform and the contribution of individual components remains poorly understood. Retrieval can be noisy or fail entirely; irrelevant passages can distract the generator; and chunking strategy, embedding model, and retrieval depth are typically tuned by intuition rather than evidence. Prior work establishes that RAG helps on knowledge-intensive tasks [1], that how passages are integrated matters [2], and surveys the full design space [3], but causal component-level ablations under a consistent evaluation protocol on the same benchmark are rare.

We address this gap with a controlled twelve-experiment study. Our key contributions are:

- A comprehensive ablation spanning retrieval family (BM25 / TF-IDF / Dense), embedding model, chunk granularity, retrieval depth k , prompt template, oracle upper bound, cross-encoder re-ranking, external distractor robustness, and generator scale.
- Statistical evidence (Holm-corrected paired tests) that the generator is the binding constraint: retrieval method, embedding model, and re-ranking are all indistinguishable within ± 0.02 EM, positively concluding they tie.
- The first clean causal estimate of when RAG hurts: pinning recall at 1 and adding hard topical distractors causes a monotone 11.5 pp EM drop ($0.673 \rightarrow 0.558$), entirely attributable to generator distraction.
- Evidence that topical retrieval misses prime parametric recall ($+0.102$ EM over an empty-context baseline, Holm $p < 10^{-4}$), and that the RAG - No-RAG gap begins to shrink as model scale grows.

2 Method

System. We implement a standard retrieve-then-read pipeline. A retriever selects the top- k passages from a corpus; these are concatenated into a prompt (question repeated before and after the context for robustness to truncation); and the generator produces a short answer. We evaluate three retrieval families: **BM25** (Okapi BM25 on a shared NFKD-normalised token stream), **TF-IDF** (TfidfVectorizer, sublinear TF, same tokeniser as BM25), and **Dense** (sentence-transformers all-MiniLM-L6-v2, 384-dim, cosine similarity). For the embedding ablation (Exp. 10) and cross-encoder re-ranking (Exp. 7) we additionally use bge-small-en-v1.5, e5-small-v2, and ms-marco-MiniLM-L-6-v2. The generator is **Flan-T5-base** [5] (≈ 250 M parameters), with greedy decoding, a 1024-token input budget, and middle-truncation to preserve the answer cue.

Dataset and corpus. We use **TriviaQA** rc [4], sampling 750 questions from the validation split (seed 42). TriviaQA was chosen because the primary references for this work (Lewis et al. [1] and Izacard and Grave [2]) both evaluate on TriviaQA, enabling direct comparison with those baselines. TriviaQA provides Wikipedia entity pages and web search hits for each question; pooling all documents across questions gives a realistic retrieval setting where the gold evidence is always present but must be located among $\sim 2,500$ documents (including 2,000 topic-agnostic Simple English Wikipedia distractors added to widen the noise floor). Documents are chunked with a fixed word-window (default 128 words, 32-word overlap). Gold answers are SQuAD-normalised (lowercase, strip articles/punctuation) and a prediction is correct if it matches any gold alias.

Evaluation. We report **Exact Match (EM)**, **Token F1**, and **Retrieval Recall@ k** (a whole-word token-span match, avoiding spurious partial-word substring hits). All metrics carry 95% percentile bootstrap CIs (1,000 resamples). For pairwise comparisons we use **McNemar’s exact test** on EM and a **paired bootstrap** on score deltas, with **Holm-Bonferroni correction** across the family of nine contrasts. Paired tests cancel question-to-question difficulty variance, providing substantially more power than comparing marginal CIs.

3 Experiments and Results

We use twelve experiments as a coordinated ablation study to answer *when retrieval helps, and when it hurts*. The core of the procedure is formed by four experiments (Exps. 5, 9, 11, 12); the remaining eight serve as **controls** that rule out alternative explanations for the core findings (Exps. 1-4, 6-8, 10). We report the core first, followed by the controls.

3.1 Core: when does retrieval help, and when does it hurt?

Does retrieval help? Massively, when parametric memory is weak (Exp. 5). RAG provides a massive benefit over the parametric floor (Table 1, top): $\Delta\text{EM} = +0.551$, Holm $p < 10^{-4}$ (discordant 419/6 questions). The No-RAG EM of 0.049 confirms that Flan-T5-base genuinely lacks these long-tail facts; the model produces confidently-shaped but incorrect entity strings (e.g., *henry viii* for “PM after Balfour”). Of 750 questions, **411 benefit**, **319 tie**, and only **6 are hurt** by retrieval; the

Table 1: Core results. All experiments use Flan-T5-base, chunk=128 words, $k=5$, instructed prompt, $n=750$ (Exp. 10: $n=400$). “Holm p ” is the McNemar p -value adjusted across the nine-contrast family; “(ref)” marks the comparison baseline.

Condition	EM	F1	R@5	Holm p
<i>RAG vs. No-RAG (Exp. 5)</i>				
No-RAG (parametric only)	0.049	0.100	—	
RAG (Dense, $k=5$)	0.600	0.668	0.865	$<10^{-4}$
<i>Retriever comparison (Exp. 1)</i>				
BM25	0.597	0.670	0.859	1.00
TF-IDF	0.603	0.673	0.852	1.00
Dense (MiniLM)	0.600	0.668	0.865	(ref)
<i>Embedding model ablation (Exp. 10)</i>				
Dense (MiniLM)	0.623	0.680	0.877	(ref)
Dense (bge-small)	0.620	0.673	0.905	—
Dense (e5-small)	0.610	0.663	0.915	—
<i>Oracle upper bound (Exp. 6)</i>				
RAG (Dense, $k=5$)	0.600	0.668	0.865	(ref)
Oracle (gold forced in)	0.619	0.684	0.999	0.0007

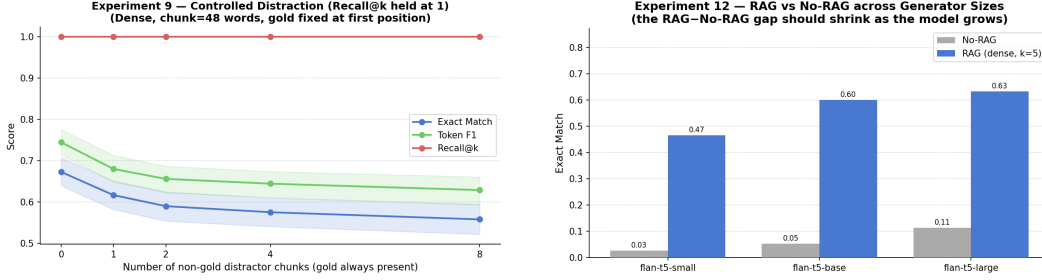


Figure 1: **Left (Exp. 9):** EM and F1 decline monotonically as hard topical distractors are added alongside a guaranteed gold chunk (Recall@ $k=1$ throughout). The $N=0$ point (gold chunk only) is the single-passage generator ceiling. **Right (Exp. 12):** RAG vs. No-RAG EM across Flan-T5 scales. The No-RAG parametric floor rises monotonically; the RAG - No-RAG gap peaks at base and begins to shrink at large, confirming that retrieval helps most when parametric memory is weakest.

6 that are hurt are questions the model already answered correctly from memory and where retrieved context distracted it. Our RAG EM of 0.600 with Dense + Flan-T5-base on TriviaQA is consistent with the 0.56–0.68 range Lewis et al. [1] report on the same dataset with DPR + BART, modulo our smaller (pooled) corpus and the lack of retriever fine-tuning.

When does it hurt? Distraction at fixed recall (Exps. 8, 9, 11). Ordinary corpus noise does not by itself create a “retrieval hurts” regime: sweeping the external distractor count (Exp. 8, {0, 500, 2000, 5000} Simple-Wikipedia articles) leaves Recall@5 and EM exactly flat across a $2.8\times$ corpus-size range, because topic-agnostic distractors never outrank gold pages. To isolate generator distraction from retrieval failure, Experiment 9 instead pins the gold chunk at rank 1 (Recall@ $k=1$ throughout) and adds $N \in \{0, 1, 2, 4, 8\}$ hard topical non-gold chunks. EM drops monotonically from 0.673 ($N=0$, single-passage ceiling) to 0.558 ($N=8$), a fall of 11.5 pp attributable entirely to distraction (Figure 1, left). Experiment 11 (forced retrieval miss, Recall@ $k=0$ by construction) closes the analysis from the other side: wrong, but topical, context scores EM 0.159, far below RAG (0.600) but well above the no-context baselines (0.049–0.057). Decomposing this jump, the instructed-prompt scaffold alone adds +0.008 EM over the bare parametric baseline, while the topical, but wrong, context adds a further +0.102 EM (Holm $p < 10^{-4}$). A retrieval miss therefore primes parametric recall rather than dragging the model below its floor.

When does the benefit fade? As the generator grows (Exp. 12). Figure 1 (right) sweeps Flan-T5 small/base/large (80M / 250M / 780M) on 150 questions. The No-RAG parametric floor rises mono-

tonically with scale (0.027→0.053→0.113), while RAG rises more slowly (0.467→0.600→0.633) towards the retrieval/generation ceiling. The RAG - No-RAG gap **peaks at base** ($\Delta = 0.547$) and **begins to shrink at large** ($\Delta = 0.520$), consistent with retrieval helping most when parametric memory is weakest. The large model is also the only one where RAG occasionally hurts (3/150 questions) as growing parametric knowledge conflicts with retrieved context.

3.2 Controls: ruling out the alternatives

Retrieval quality is not the binding constraint (Exps. 1, 6, 7, 10). Four control experiments converge on the same conclusion: the generator, not the retriever, limits performance. First, the three retrieval families are statistically indistinguishable (Table 1, middle; Holm $p=1.0$ for all three pairs): Dense achieves the highest Recall@5 (0.865) but not the highest EM, so the residual error is generator-side. Second, the oracle upper bound (Exp. 6, Table 1, bottom), which forces the gold chunk in for the 13% of questions Dense misses, yields only +0.019 EM (Holm $p=0.0007$) - small but reliable. Third, cross-encoder re-ranking (Exp. 7, *ms-marco-MiniLM-L-6-v2*, top-50→top-5) lifts Recall@5 by 0.017 but its EM gain (+0.017, Holm $p=0.73$) is not significant. Fourth, the embedding-model ablation (Exp. 10, $n=400$) shows the same dissociation: e5-small reaches the highest Recall@5 (0.915) yet the lowest EM (0.610), while MiniLM has the lowest Recall@5 (0.877) but the highest EM (0.623). A stronger retriever finds more gold chunks; the generator extracts no better answers from them.

Pipeline hyperparameters: chunk size, depth, and prompt (Exps. 2, 3, 4). At a fixed 1024-token budget (Exp. 2), smaller chunks win: 64-word chunks (≈ 8 packed per prompt) achieve EM 0.596 vs. 0.517 for 512-word chunks (≈ 1.2 per prompt), since large chunks risk losing the entire context window on a single retrieval miss. Retrieval depth k (Exp. 3, chunk=48 words) improves monotonically through $k=10$ (EM: 0.433→0.583) with no distraction penalty, consistent with distraction only materialising once the answer is already guaranteed present (Exp. 9). Prompt template (Exp. 4) has no significant effect: concise vs. instructed differ by only 0.016 EM (Holm $p=0.18$).

4 Conclusion

We investigated utility of RAG through a controlled ablation of a pipeline on TriviaQA. Four contrasts survived Holm correction: RAG vs. No-RAG (+0.551 EM, the dominant effect); both comparisons of forced-miss against no-context baselines (+0.102–0.110 EM, topical priming); and Oracle vs. RAG (+0.019 EM, a small, but reliable retrieval-failure component first resolved at $n=750$). Retrieval method, embedding model, and reranking are all Holm-non-significant within ± 0.02 EM with large discordant counts, supporting the idea that these factors have little influence on RAG success. Experiment 9 demonstrates a clear case of RAG suboptimality: each additional topical distractor reduces EM monotonically, even when the answer is guaranteed to be present. Overall, the results suggest that for small instruction-tuned generators, effort is better invested into the generator than into retrieval performance.

Limitations. The study uses small encoder-decoders (up to 780M parameters) on a much smaller and less adversarial corpus than open-domain Wikipedia. A DPR model fine-tuned on TriviaQA would likely widen the sparse-dense gap currently observed as a tie. Much larger generators (LLaMA-class) would raise the parametric floor further and shrink the RAG benefit; multi-hop questions and adversarially constructed distractors are left to future work.

References

- [1] Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., ... Kiela, D. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *NeurIPS 2020*.
- [2] Izacard, G., & Grave, E. (2021). Leveraging Passage Retrieval with Generative Models for Open Domain Question Answering. *EACL 2021*.
- [3] Gao, Y., Xiong, Y., Gao, X., Jia, K., Pan, J., Bi, Y., ... Wang, H. (2024). Retrieval-Augmented Generation for Large Language Models: A Survey. *arXiv:2312.10997*.

- [4] Joshi, M., Choi, E., Weld, D., & Zettlemoyer, L. (2017). TriviaQA: A Large Scale Distantly Supervised Challenge Dataset for Reading Comprehension. *ACL 2017*.
- [5] Chung, H. W., Hou, L., Longpre, S., Zoph, B., Tay, Y., Fedus, W., . . . Wei, J. (2022). Scaling Instruction-Finetuned Language Models. *arXiv:2210.11416*.
- [6] Robertson, S., & Zaragoza, H. (2009). The Probabilistic Relevance Framework: BM25 and Beyond. *Foundations and Trends in Information Retrieval*, 3(4).

A Additional Figures

Figures 2–16 show all fifteen experiment plots from the report-scale run ($n=750$, $n=400$ for Exp. 10, $n=150$ for Exp. 12) on GPU (HPC). Error bars are 95% percentile bootstrap CIs (1,000 resamples).

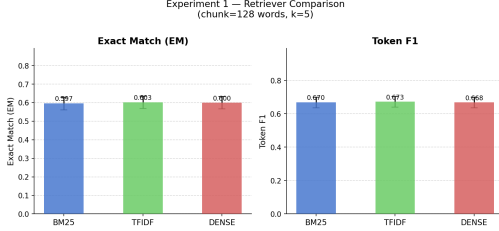


Figure 2: Exp. 1 — Retriever comparison (BM25 / TF-IDF / Dense). All three are statistically indistinguishable (Holm $p=1.0$ for all pairs).

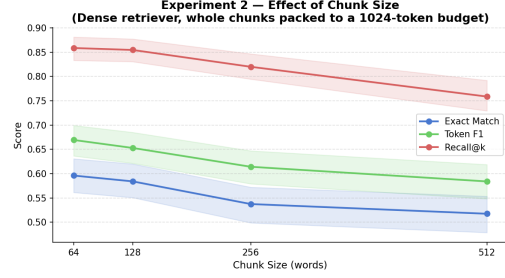


Figure 3: Exp. 2 — Chunk-size ablation at a fixed 1024-token budget. Smaller chunks pack more passages and achieve higher recall and EM.

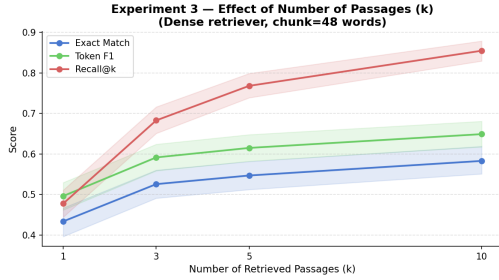


Figure 4: Exp. 3 — Retrieval depth sweep ($k \in \{1, 3, 5, 10\}$, chunk=48 words). EM and Recall@ k both increase monotonically; no distraction penalty appears at standard retrieval.

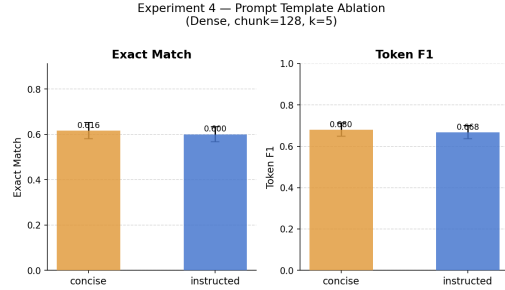


Figure 5: Exp. 4 — Prompt template: concise vs. instructed. Difference is 0.016 EM (Holm $p=0.18$, non-significant).

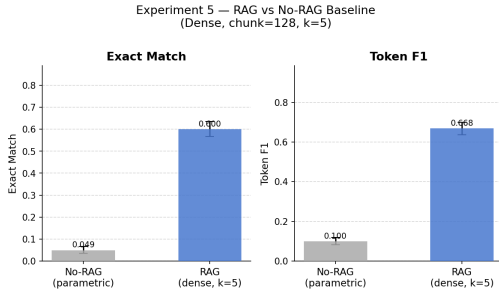


Figure 6: Exp. 5 — RAG vs. No-RAG. $\Delta\text{EM} = +0.551$ (Holm $p < 10^{-4}$; 411 helps / 6 hurts / 319 ties out of 750 questions).

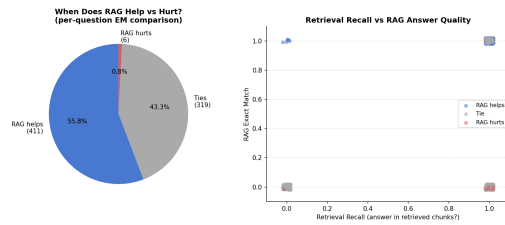


Figure 7: Exp. 5 error analysis: per-question RAG helps / hurts / ties breakdown and scatter of Recall@5 vs. ΔEM .

B Complete Results Tables

C Statistical Significance Tests

Table 8 reports all nine pairwise contrasts with raw and Holm-Bonferroni-adjusted p -values. Four contrasts survive correction (bold): RAG vs. No-RAG, both Forced-Miss contrasts, and Oracle vs. RAG.

Experiment 5 — Qualitative Examples: When RAG Helps vs Hurts

Category	Question (Instruction)	Model	Relevant point	Not point
RAG helps	What general name is given to a writing?	ReLSTM	di	Author
RAG helps	What Gilbert and Sullivan operates in in	AutoGPTQ	symphony	burlesque
RAG helps	Who is the patron saint of dancers?	St-Blue	Saint John	Blas
RAG hurts	How many Thengles are there on the flag	flair	four	22
RAG hurts	Scientist Ernest Rutherford once said ab	Phrasit	phrasit	phrasit or dense collect
RAG hurts	A number of which professions would not	Eligapher	Holographer	Antennemaster

Figure 8: Exp. 5 qualitative examples where RAG helps and where it hurts.

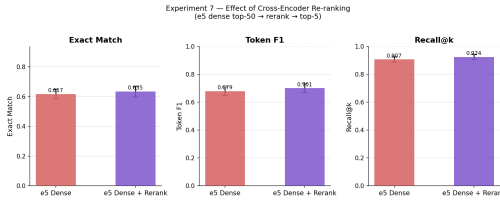


Figure 10: Exp. 7 — Cross-encoder re-ranking (e5 first stage, top-50→top-5). Recall@5 rises by 0.017 but EM gain (+0.017) is not significant (Holm $p=0.73$).

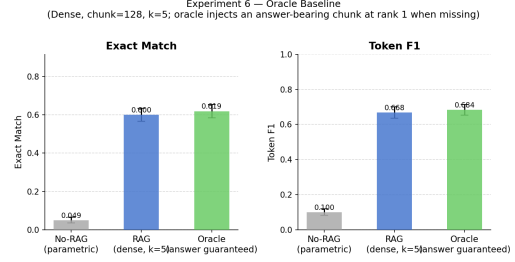


Figure 9: Exp. 6 — Oracle vs. RAG vs. No-RAG. Oracle gain: +0.019 EM (Holm $p=0.0007$), quantifying the small reliable retrieval-failure component.

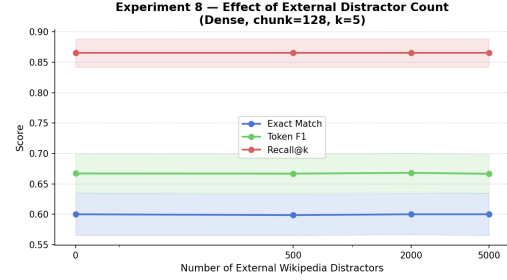


Figure 11: Exp. 8 — External distractor count sweep ($\{0, 500, 2000, 5000\}$ Simple-Wikipedia articles). Recall@5 and EM are exactly flat across a $2.8\times$ corpus-size range: topic-agnostic distractors never out-rank gold pages.

D Implementation Details

Chunking. Fixed word-window chunking with 32-word overlap. For a document of W words and chunk size c : chunk i spans words $[i(c-32), i(c-32)+c]$.

Generation. Flan-T5-base with greedy decoding (`do_sample=False`, `num_beams=1`, `max_new_tokens=32`). Long prompts are *middle-truncated*: prefix (question) and suffix (answer cue) are preserved; only the context body is trimmed. All generation is cached to `data/cache/generation_cache.json` keyed by MD5 of `model_name + rendered_prompt`, so different generator sizes (Exp. 12) never collide.

Dense retrieval. Query and passage embeddings are L2-normalised; retrieval is by dot product (equivalent to cosine similarity). Passage embeddings are cached per chunk-size to avoid re-encoding across experiments. For Exp. 10, each model receives its recommended asymmetric query/passage instruction prefixes (e5: "query: "/"passage: "; bge: search instruction on the query side; MiniLM: no prefix).

Reproducibility. All random processes use fixed seeds (corpus sampling seed 42, bootstrap seed 42). Running the notebook twice produces identical results. Expected end-to-end runtime at $n=750$: approximately 4–8 h on CPU (significantly less on GPU/MPS with embedding caching).

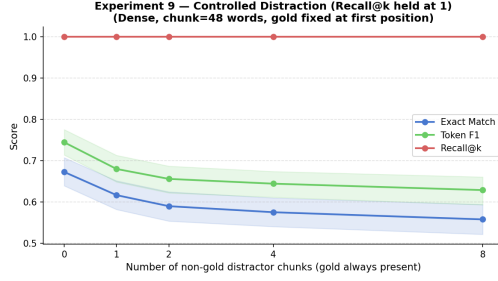


Figure 12: Exp. 9 — Controlled distraction (Recall@ $k = 1$ throughout). Adding hard topical distractors drives EM monotonically from 0.673 ($N = 0$) to 0.558 ($N = 8$).

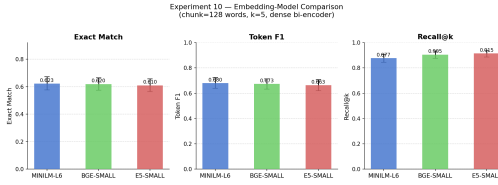


Figure 14: Exp. 10 — Embedding model ablation ($n = 400$). e5-small achieves the highest Recall@5 (0.915) yet the lowest EM (0.610); MiniLM has the lowest recall but the highest EM (0.623).

Paired Significance Tests
(same questions; McNemar on EM, paired bootstrap on deltas; Holm-Bonferroni adjusted p alongside raw)

Contrast (A - B)	EM	McNemar p	EM Δ (std)	EM Δ (std)	EM Δ (std)	EM Δ (std)	EM Δ (std)	EM Δ (std)	EM Δ (std)	EM Δ (std)
EM12 - Dense	0.673	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
EM12 - No-RAG	0.673	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
EM12 - T5-27	0.673	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
EM12 - T5-27	0.673	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
EM12 - T5-27	0.673	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
EM12 - T5-27	0.673	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
EM12 - T5-27	0.673	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
EM12 - T5-27	0.673	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
EM12 - T5-27	0.673	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
EM12 - T5-27	0.673	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Figure 13: Paired significance table (McNemar on EM, paired bootstrap on deltas; raw and Holm-adjusted p -values for all nine contrasts).

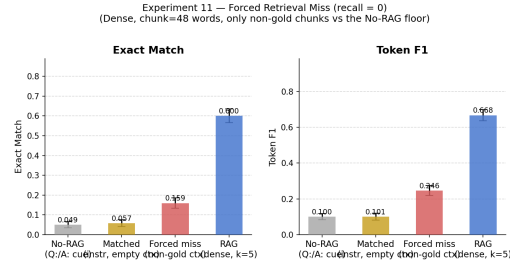


Figure 15: Exp. 11 — Forced retrieval miss (Recall@ $k = 0$). Topical-but-wrong context adds +0.102 EM over the empty-context baseline (Holm $p < 10^{-4}$) via parametric priming.

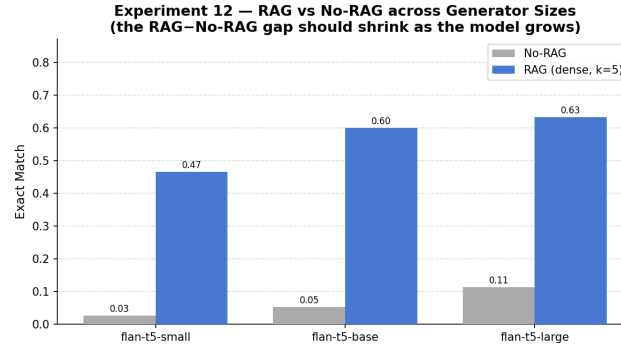


Figure 16: Exp. 12 — Generator-scale sweep ($n = 150$). No-RAG floor rises with scale (0.027→0.053→0.113); RAG–No-RAG gap peaks at base ($\Delta = 0.547$) and shrinks at large ($\Delta = 0.520$).

Table 2: Exp. 2 — Chunk size at fixed 1024-token budget (Dense, $n = 750$, 30 candidates packed greedily).

Chunk (words)	Mean packed k	EM	Token F1	Recall@ k
64	8.27	0.596	0.669	0.859
128	4.29	0.584	0.653	0.855
256	2.14	0.537	0.614	0.820
512	1.19	0.517	0.584	0.759

Table 3: Exp. 3 — Retrieval depth k (Dense, chunk = 48 words, $n = 750$; full range fits 1024-token budget untruncated).

k	EM	Token F1	Recall@ k
1	0.433	0.496	0.477
3	0.525	0.591	0.683
5	0.547	0.615	0.768
10	0.583	0.649	0.855

Table 4: Exp. 9 — Controlled distraction (Recall@ $k = 1$ by construction; gold chunk pinned at rank 1, $n \approx 750$, chunk = 48 words).

N distractors	EM	Token F1	Recall@ k
0	0.673	0.745	1.00
1	0.617	0.680	1.00
2	0.590	0.656	1.00
4	0.575	0.645	1.00
8	0.558	0.629	1.00

Table 5: Exp. 11 — Forced retrieval miss ($n \approx 750$, chunk = 48 words; Recall@ $k = 0$ by construction for the forced-miss arm).

Condition	EM	Token F1	Recall@ k
No-RAG (bare Q: /A: cue)	0.049	0.100	—
Matched (instructed prompt, <i>empty</i> context)	0.057	0.101	—
Forced miss (instructed prompt, non-gold chunks)	0.159	0.246	0.00
RAG (Dense, $k = 5$, Exp. 5)	0.600	0.668	0.865

Table 6: Exp. 12 — Generator-scale sweep ($n = 150$; trend demonstration on a representative subset).

Generator	No-RAG EM	RAG EM	Δ EM	Helps / Hurts / Ties
flan-t5-small (≈ 80 M)	0.027	0.467	+0.440	67 / 1 / 82
flan-t5-base (≈ 250 M)	0.053	0.600	+0.547	83 / 1 / 66
flan-t5-large (≈ 780 M)	0.113	0.633	+0.520	81 / 3 / 66

Table 7: Exp. 8 — External distractor count sweep (Dense, chunk = 128, $k = 5$, $n = 750$; corpus sizes are approximate).

Distractors	Corpus docs	EM	Token F1	Recall@5
0	$\sim 2,800$	0.600	0.667	0.865
500	$\sim 3,300$	0.599	0.667	0.865
2000	$\sim 4,800$	0.600	0.668	0.865
5000	$\sim 7,800$	0.600	0.667	0.865

Table 8: Paired significance results ($n = 750$). McNemar’s exact test on EM; paired bootstrap on F1 deltas (10,000 resamples). Holm–Bonferroni correction across all nine contrasts. Discordant counts are (A better / B better).

Contrast (A – B)	Δ EM	Disc.	McN raw	McN Holm	Δ F1	95% CI	F1 Holm
BM25 – Dense	–0.003	70/72	0.933	1.000	+0.002	[–0.026, +0.030]	1.000
TF-IDF – Dense	+0.003	67/65	0.931	1.000	+0.005	[–0.021, +0.032]	1.000
BM25 – TF-IDF	–0.005	38/42	0.738	1.000	–0.003	[–0.025, +0.018]	1.000
Instructed – Concise	–0.016	8/20	0.036	0.179	–0.012	[–0.025, +0.001]	0.312
RAG – No-RAG	+0.551	419/6	< 0.001	< 0.001	+0.568	[+0.534, +0.603]	< 0.001
Dense+Rerank – e5-Dense	+0.017	47/34	0.182	0.728	+0.022	[–0.001, +0.044]	0.312
Oracle – RAG	+0.019	14/0	0.0001	0.0007	+0.016	[+0.007, +0.026]	0.001
ForcedMiss – No-RAG	+0.110	104/23	< 0.001	< 0.001	+0.148	[+0.118, +0.179]	< 0.001
ForcedMiss – InstrNoCtx	+0.102	104/29	< 0.001	< 0.001	+0.145	[+0.114, +0.177]	< 0.001